

Noise-Tolerant Multilabel Feature Selection Based on Sample Reliability Degree and Fuzzy Association Confidence

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Abstract—Multilabel feature selection (MFS) has received much attention because it can effectively mitigate the impact of “dimensionality curse.” Ambiguity and unavoidable label noise in real life pose significant challenges to MFS. So far, there are relatively few studies on MFS considering label noise. Existing few studies do not consider that label sets of different samples are affected by noise to different degrees, and ignoring differences between samples may reduce the robustness of MFS. Moreover, they explore the fuzzy relationship between samples and labels from a local perspective, which is not comprehensive. In addition, they lack a monotonic metric for measuring attribute importance, leading to the unreasonable selected feature subset. To solve these issues, we propose a noise-tolerant MFS method by monotonic noise-tolerant fuzzy confidence conditional entropy with sample reliability degree and fuzzy association confidence called SRDAC. First, sample reliability degree is defined from the perspective of consistency to distinguish samples affected by different levels of label noise. Second, the idea of global similarity and the fuzzy association confidence between samples and labels are proposed, and corresponding noise-tolerant label enhancement for different types of samples is carried out to better explore the relationship between samples and labels. Third, based on noise-tolerant label space, the noise-tolerant fuzzy confidence conditional entropy is proposed to more reasonably measure the relationship between attributes. Finally, a robust MFS method based on monotonic noise-tolerant fuzzy confidence conditional entropy is proposed. Extensive experiments show that the effectiveness and robustness of SRDAC is better than other representative methods.

Index Terms—Fuzzy association confidence, fuzzy rough sets, label noise, multilabel feature selection (MFS), sample reliability degree (SRD).

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I. INTRODUCTION

IN RECENT years, multilabel learning has garnered significant attention in practical scenarios including gene expression prediction [1], audio and video classification [2], and image annotation [3]. In contrast to single-label supervised learning, where each sample is associated with a single class label, multilabel learning enables the simultaneous assignment of multiple labels to a single sample [4]. For example, an image can be annotated with multiple semantics, including clouds, sky, sea, and sun [5]. However, multilabel learning tasks often suffer from “dimensionality curse,” which increase the computational burden [6]. Therefore, reducing the dimensionality of multilabel data is crucial to improve the performance of multilabel learning [7]. As an effective dimensionality reduction method, multilabel feature selection (MFS) aims to select the most relevant feature subset. Moreover, the MFS method retains the physical meaning of selected features [8]. Therefore, MFS has become an indispensable preprocessing step [9].

According to different selection strategies, MFS methods can be summarized into three categories: filter, wrapper, and embedded methods [10]. Filter methods select the optimal feature subset [11] by the method independent of any classifier and evaluate the features according to various metrics [12]. Wrapper methods rely on a specific classifier and use the training accuracy of a predetermined learning algorithm to evaluate features [7]. Embedded methods [13] integrate feature selection into the learning model construction, and obtain the optimal feature subset by optimizing the objective function. In contrast to wrapper methods and embedded methods, filter methods possess good interpretability and less risk of overfitting. Therefore, this article will concentrate on the filter methods.

Information theory can effectively measure linear and nonlinear relationships between random variables [14]. Fuzzy rough sets provide an effective method to deal with uncertainty and imprecise information. Therefore, information theory and fuzzy rough sets have been introduced into MFS around filter methods. For example, Qian et al. [15] combined random variable distribution with granular computing and proposed an MFS method for label distribution by mutual information and label enhancement. Che et al. [16] proposed an MFS method that considers label correlation by using fuzzy rough sets. Lin et al. [17] defined three different neighborhood mutual information and designed three MFS methods. Sun et al. [18] investigated the concept

of edge-based fuzzy neighborhood granularity and developed a multilabel oriented fuzzy neighborhood rough set model that integrates the principles of fuzzy rough sets and neighborhood rough sets into a unified framework. Xiong et al. [19] designed an MFS method by label distribution and fuzzy mutual information to assess the feature importance. Yin et al. [20] realized multi-scale cognition by analyzing and fusing knowledge granularity at different scales, and handled missing labels by reconstructing the label space. Dai et al. [21] designed an MFS method that considers two new relationships between features and labels by combining label mutual assistance and discriminant function.

Although these methods have achieved good results in MFS, they do not perform well when dealing with multilabel data with label noise. Having satisfactory classification performance often requires a highly accurate label space. However, the data within practical scenarios often produce a certain number of false labels during the labeling process due to the unavoidable errors and the ambiguity of multilabel space, which leads to the existence of noisy labels. Classical MFS cannot distinguish which labels are more reliable, thus treating all labels equally. When the label set contains noise, these methods may incorrectly rely on the noisy labels for MFS, leading to an amplification of the influence of the noisy labels on MFS. Therefore, label noise will interfere with the classical MFS methods to evaluate the importance of features and affect the selection of feature subsets, which brings great challenges to MFS. So far, there are relatively few studies on MFS considering label noise. Recently, Yin et al. [22] designed a robust MFS based on multineighborhood fuzzy β covering granular space that considers feature multicorrelations. Yin et al. [23] proposed a new fuzzy neighborhood rough set model and designed the corresponding antinoise heuristic MFS method NRFSEFN. Yin et al. [24] designed a robust MFS method GRMFS with fuzzy dependencies and feature interactions based on graph structure to process multilabel data containing noise.

However, these methods still have certain limitations, which drives us to address these challenges. Therefore, the motivations of this study are as follows. First, the influence degree of label noise on label sets of different samples varies, and ignoring the differences between samples in these methods will affect the robustness of MFS. Therefore, sample reliability degree (SRD) is defined from the perspective of consistency between feature space and label space, so as to reflect the degree to which different samples are affected by noise. Second, exploring the fuzzy relationship between samples and labels from a local perspective is not comprehensive. Unlike previous methods that only consider similarity in the feature space, this article proposes the idea of global similarity that comprehensively considers feature similarity and label similarity based on the adjustment of SRD. At the same time, the fuzzy association confidence measure between samples and labels is further proposed to carry out corresponding noise-tolerant label enhancement for different types of samples. The fuzzy relationship between samples and labels is investigated in depth to obtain more information hidden in the label space, reduce the interference of label noise on MFS, and enhance its robustness. Third, the lack of a monotonic metric for measuring the importance of attributes leads to the selected feature subset easily disturbed by noise. Measuring the

amount of information contained in attributes is an important issue. To solve this issue, after obtaining the noise-tolerant label space, we define the noise-tolerant fuzzy confidence entropy and noise-tolerant fuzzy confidence conditional entropy to measure the uncertainty of attributes, and propose a robust MFS method based on monotonic noise-tolerant fuzzy confidence conditional entropy. The main contributions of this article are summarized as follows.

- 1) Aiming at the problem that the label sets of different samples are affected by noise to different degrees, the similar sample set in the feature space is defined. On this basis, from the point of view that the feature and label space have consistency, the SRD is proposed to reflect the degree of the label set of the samples affected by noise, and the samples are classified into two categories of credible samples and suspicious samples according to the reliability degree of the samples.
- 2) To obtain a more accurate relationship between samples and labels, based on the SRD, fuzzy association confidence measure between samples and labels is proposed from the perspective of global similarity. Corresponding noise-tolerant label enhancement for different types of samples is carried out to convert the noisy logical values into noise-tolerant confidence label values, which mines more classification information implicit in the label space.
- 3) To quantify the decision uncertainty of fuzzy granularity in multilabel data containing label noise, based on the SRD and the noise-tolerant label space, the noise-tolerant fuzzy confidence entropy uncertainty measure is defined, and the related properties are investigated.
- 4) Based on monotonic noise-tolerant fuzzy confidence conditional entropy, a novel robust MFS method is proposed to measure the importance of features, and the effectiveness and robustness of the proposed method are verified through a series of experiments.

The rest of this article is organized as follows. Section II reviews some fundamental concepts. Section III describes the details of the proposed robust MFS method. In Section IV, the proposed method is compared with other representative MFS methods. Finally, Section V concludes this article.

II. PRELIMINARIES

In this section, we briefly review some fundamental concepts of fuzzy similarity relation and label weight.

A. Fuzzy Similarity Relation

Fuzzy similarity relation is usually defined as a measure between two fuzzy sets, which can be used to calculate the similarity between samples. This is very useful when dealing with uncertainty, imprecision, or subjective judgment. The definition of fuzzy similarity relation is as follows.

Definition 1 (See [25] [26]): Given a multilabel information table $MLT = \langle U, C \cup L, V, F \rangle$, $U = \{x_1, x_2, \dots, x_n\}$ is a sample set, $C = \{f_1, f_2, \dots, f_m\}$ is a feature set, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, $V = \bigcup_{a \in C \cup L} V_a$ is the domain of attributes, and $F : U \times \{C \cup L\} \rightarrow V$ is a map function. For

any $x_i, x_j \in U, B \subseteq C, R_B(x_i, x_j)$ denotes the fuzzy similarity between x_i and x_j , the fuzzy relation R_B is called a fuzzy similarity relation if R_B satisfies:

- 1) *Reflexivity*: $R_B(x_i, x_i) = 1$;
- 2) *Symmetry*: $R_B(x_i, x_j) = R_B(x_j, x_i)$.

Definition 2 (See [27] [28]): R_B represents fuzzy similarity relation induced by B , then relation matrix $M(R_B)$ is presented as follows:

$$M(R_B) = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{pmatrix} \quad (1)$$

where $r_{ij} = R_B(x_i, x_j) = \min_{b \in B} R_b(x_i, x_j)$ can be utilized to measure the fuzzy similarity between two samples x_i and x_j . In this article, $R_b(x_i, x_j) = \max(\min(\frac{b(x_i) - b(x_j) + \sigma}{\sigma}, \frac{b(x_j) - b(x_i) + \sigma}{\sigma}), 0)$, where σ denotes the standard deviation of feature b and $r_{ij} \in [0, 1]$.

Definition 3 (See [29]): Suppose $B \subseteq C, R_B$ denotes a fuzzy similarity relation induced by B . The nonempty set U can be divided into the following granularity:

$$U/R_B = \{[x_i]_{R_B}\}_{i=1}^n \quad (2)$$

where $[x_i]_{R_B} = \frac{r_{i1}}{x_1} + \frac{r_{i2}}{x_2} + \cdots + \frac{r_{in}}{x_n}$ represents the fuzzy similar class of x_i generated by R_B .

B. Label Weight

In multilabel data, each instance is associated with multiple labels and there is often some correlation between labels. Mutual information can be utilized to calculate the correlation between labels, which reflects the degree of correlation between labels. In order to measure the importance of each label, mutual information is used to calculate the weight of each label based on the construction of weighted undirected graphs between labels.

Definition 4: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $U = \{x_1, x_2, \dots, x_n\}$ is a sample set, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, $l_{k_1}, l_{k_2} \in L$, the correlation between two labels is calculated using mutual information as follows:

$$w(l_{k_1}, l_{k_2}) = \sum_{a \in \{0,1\}} \sum_{b \in \{0,1\}} P(l_{k_1} = a, l_{k_2} = b) * \log \frac{P(l_{k_1} = a, l_{k_2} = b)}{P(l_{k_1} = a)P(l_{k_2} = b)} \quad (3)$$

where P represents the proportion of samples satisfying the corresponding conditions in all samples.

Definition 5 (See [30]): Given a label set $L = \{l_1, l_2, \dots, l_q\}$, we get the final label weights by updating the label weights once using the following formula:

$$LW(l_{k_1}) = (1 - d) + d \sum_{l_{k_2} \in L - \{l_{k_1}\}} \frac{LW(l_{k_2})w(l_{k_1}, l_{k_2})}{SW(l_{k_2})} \quad (4)$$

where

$$SW(l_{k_2}) = \sum_{l_{k_1} \in L - \{l_{k_2}\}} w(l_{k_1}, l_{k_2}) \quad (5)$$

where $LW(l_{k_1})$ and $LW(l_{k_2})$ denote the weights of labels l_{k_1} and l_{k_2} , respectively; $w(l_{k_1}, l_{k_2})$ indicates the degree of correlation between l_{k_1} and l_{k_2} ; $SW(l_{k_2})$ represents the total correlation degree of all edges originating from label node l_{k_2} ; d denotes the damping factor, which is usually set to 0.85. The initial weight of all labels is set to $1/q$, where q denotes the total number of labels.

III. PROPOSED ROBUST MFS METHOD

In this section, the proposed robust MFS method is introduced in detail.

A. Sample Reliability Degree and Sample Classification

In a particular dataset, adjacent data points or feature values are likely expected to have similar outputs or labels. In other words, in a multilabel information table, samples that are similar in the feature space are likely expected to have a similar degree of similarity in the label space. When the label space is contaminated by noise, a sample that should have been more similar to its adjacent samples has a lower degree of similarity under the contaminated label space, and therefore, the confidence level of the label set associated with that sample is also reduced.

In multilabel environment, each sample is associated with multiple labels, and noise may appear on any one or several labels. The number of noisy labels contained in each sample is unknown. In this case, if the differences between samples are ignored, the MFS method is unable to distinguish which samples have more reliable labels, thus treating all samples equally. Then when the label sets of some samples contain more noise, the method may incorrectly rely on these noisy labels for feature selection, which reduces the robustness of MFS. In order to solve this problem, a new metric called the SRD is defined, which reflects the degree of credibility of the labels associated with a sample when the label space may be affected by noise.

Definition 6: Each sample has a set of k samples that are most similar to it in the feature space, and this set is defined as a similar sample set. The detailed definition is as follows:

$$sim(x_i) = \{x_j | R_C(x_i, x_j) \geq R_C(x_i, x_{ik}) \wedge (j \neq i)\} \quad (6)$$

where $R_C(x_i, x_j)$ denotes the fuzzy similarity between sample x_i and sample x_j under the feature space. x_{ik} represents the k th sample in descending order of fuzzy similarity between x_i and the rest of the samples.

The label set associated with each sample is compared separately with the label set of the samples in its similar sample set. The proportion of the same labels between a sample and the samples in its similar sample set is called the SRD.

Definition 7: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, q indicates the number of labels, $sim(x_i)$ denotes similar sample set of the sample x_i , then the SRD of the sample x_i is defined as

$$SRD(x_i) = \frac{\sum_{x_j \in sim(x_i)} \sum_{l_k \in L} (l_k^{x_i} = l_k^{x_j})}{|sim(x_i)| * q} \quad (7)$$

where $l_k^{x_i}$ represents the value of sample x_i on label l_k , $|sim(x_i)|$ represents the number of elements.

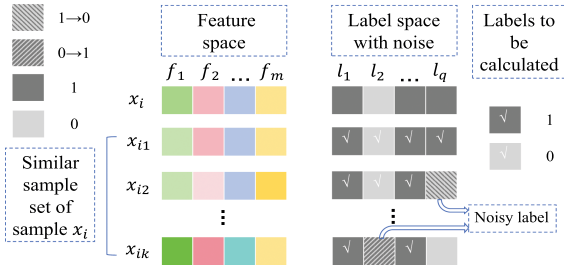


Fig. 1. Diagram of SRD.

Proposition 1. (Nonnegative): $SRD(x_i) \geq 0$.

Proof: This proof is straightforward. $\square$

Fig. 1 shows a diagram of SRD, where shaded blocks are labels that may contain noise. The greater the proportion of the same label in a sample with its similar sample set, the higher the reliability degree of the sample. This indicates that the label set of this sample is less affected by noise, that the label set is closer to the real label set, and that the similarity between this sample and other samples in the label space has greater information value.

In order to distinguish samples with different confidence levels and attenuate the interference of label noise on the feature selection process, all samples are classified based on their SRD. Due to the different label situations in different data, the parameter λ is introduced as the confidence threshold to divide the credible samples and the suspicious samples.

Definition 8: $SRD(x_i)$ represents the reliability degree of the sample x_i , λ is the confidence threshold, then the detailed definitions of the credible sample set HC^λ and the suspicious sample set LC^λ are as follows:

$$HC^\lambda = \{x_i | SRD(x_i) \geq \mathcal{T}_\lambda\} \quad (8)$$

$$LC^\lambda = \{x_i | SRD(x_i) < \mathcal{T}_\lambda\} \quad (9)$$

where the SRD values of all samples are sorted in descending order and $\mathcal{T}_\lambda$ represents the reliability degree of the λ th sample.

When the reliability degree of the sample is greater than $\mathcal{T}_\lambda$, the similarity of the feature space is more consistent with the similarity of the label space, which is called a credible sample. When the reliability degree of the sample is less than $\mathcal{T}_\lambda$, the similarity of the feature space is more different from the similarity of the label space, which is called a suspicious sample. Fig. 2 shows a diagram of the different types of samples. It can be seen from the figure that only a small number of samples in the similar sample set of suspicious sample x_i with low sample reliability also have similarity close to the feature similarity in the label space.

B. Fuzzy Association Confidence Measure

Since each label contributes differently to the data, label weights are combined to compute the similarity relation between samples under the original label space.

Definition 9: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $L = \{l_1, l_2, \dots, l_q\}$ is a label set. $LW(l_k)$ represents the importance

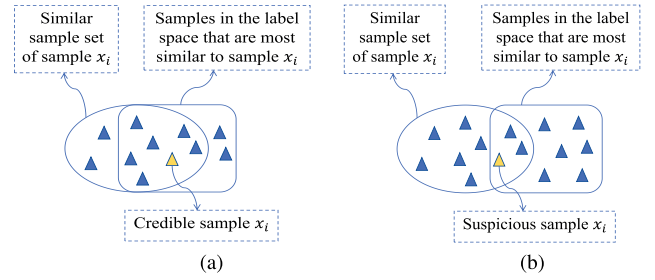


Fig. 2. Diagram of different types of samples. (a) Credible sample. (b) Suspicious sample.

of label l_k , then the similarity under the entire label set L is defined as follows:

$$R_L(x_i, x_j) = \sum_{k=1}^q LW(l_k) R_{l_k}(x_i, x_j) \quad (10)$$

where

$$R_{l_k}(x_i, x_j) = \begin{cases} 1, & l_k^{x_i} = l_k^{x_j} \\ 0, & l_k^{x_i} \neq l_k^{x_j} \end{cases} \quad (11)$$

where $l_k^{x_i}$ represents the value of sample x_i on label l_k .

In practical scenarios, multiple associated semantics of objects often exhibit differences. At the same time, due to the inevitable noise in the label space, the hypothesis of uniform distribution in multilabel learning cannot accurately suit these real-world situations, which may lead to the loss of generalization ability. To further explore the association between samples and labels, the fuzzy association confidence measure between samples and labels is proposed from the perspective of global similarity, i.e., considering both feature similarity and label similarity between samples. The fuzzy association confidence measure between the sample and the label is used to perform corresponding noise-tolerant label enhancement for different types of samples divided according to the SRD. Among them, the perspective of global similarity is embodied in the numerator of fuzzy association confidence measure. The proportion of label similarity in global similarity is adjusted by SRD to attenuate the interference of label noise on the feature selection.

For a credible sample, the proportion of the same labels between it and the samples in its similar sample set is larger, which indicates that the label set of the sample is weakly affected by noise and the reliability of the overall label set is higher. Therefore, it is essential to retain the original label information of these samples as much as possible to avoid introducing unnecessary noise and errors. For this reason, the original label values are kept unadjusted for unassigned label l_k ($l_k = 0$).

Definition 10: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $x_i \in HC^\lambda$, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, the credible negative fuzzy association confidence between credible sample x_i and unassigned label l_k is defined as

$$AC_{cn}^\lambda(x_i, l_k) = 0. \quad (12)$$

For associated label l_k ($l_k = 1$), in order to more accurately depict the relative importance of each label to sample, the

credible positive fuzzy association confidence between credible samples and associated labels is defined by comprehensively considering the global similarity between the target sample and samples with the same label in its similar sample set to carry out appropriate transformation. The similarity under label space is calculated by the similarity of the entire label set.

Definition 11: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $x_i \in HC^\lambda$, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, $SRD(x_i)$ indicates the reliability degree of the sample x_i , $sim(x_i)$ denotes similar sample set of the sample x_i , the credible positive fuzzy association confidence between credible sample x_i and associated label l_k is defined as

$$AC_{cp}^\lambda(x_i, l_k) = \frac{\sum_{\substack{x_t \in sim(x_i) \\ l_k^x = 1}} [R_C(x_i, x_t) + SRD(x_i) * R_L(x_i, x_t)]}{\sum_{x_t \in sim(x_i)} [R_C(x_i, x_t) + SRD(x_i) * R_L(x_i, x_t)]} \quad (13)$$

where $R_C(x_i, x_t)$ denotes the similarity between sample x_i and x_t in its similar sample set under the feature space, R_L indicates the similarity between sample x_i and x_t in its similar sample set under the entire label set L .

The proportion of similarity in label space increases with the increase in the reliability degree of the sample. The less noise the sample contains, the more reliable monitoring information we can use in the label space.

For a suspicious sample, the proportion of same labels between it and the samples in its similar sample set is smaller. This indicates that the label set of this sample is seriously affected by noise and the reliability of the overall label set is reduced. In the process of label enhancement for suspicious samples, using the similarity under the entire label set is easy to make the performance of MFS method disturbed by noise. Therefore, the similarity of the label space in the global similarity is reflected by the similarity under the current label combined with the reliability degree of the corresponding sample.

For unassigned labels, the effect of label noise is attenuated by calculating the average of the global similarity between the sample and the samples in its similar sample set, where feature similarity can be used to maintain basic correlation information.

Definition 12: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $x_i \in LC^\lambda$, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, $SRD(x_i)$ indicates the reliability degree of the sample x_i , $sim(x_i)$ denotes a similar sample set of the sample x_i , the suspicious negative fuzzy association confidence between suspicious sample x_i and unassigned label l_k is defined as

$$AC_{sn}^\lambda(x_i, l_k) = \frac{\sum_{x_t \in sim(x_i)} [R_C(x_i, x_t) + SRD(x_i) * R_{l_k}(x_i, x_t)]}{|sim(x_i)|}. \quad (14)$$

The reliability degree of the sample can adjust the proportion of the similarity of label space in the global similarity to attenuate the interference caused by label noise. When the label space of the sample contains more noise, the reliability degree of the sample is smaller, and the reliable supervisory information

provided by the label set of the sample is reduced. In this case, reducing the proportion of the similarity of the label space in the global similarity can improve the accuracy of the global similarity and achieve more accurate label enhancement.

For associated labels of suspicious sample x_i , solving in the same way as for unassigned labels leads to excessive enhancement of the suspicious association label. Therefore, the global similarity is further defined as the product of feature and label similarity, no longer retaining the basic correlation information, which transforms towards the majority of labels as much as possible.

Definition 13: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $x_i \in LC^\lambda$, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, $SRD(x_i)$ indicates the reliability degree of the sample x_i , $sim(x_i)$ denotes a similar sample set of the sample x_i , the suspicious positive fuzzy association confidence between suspicious sample x_i and associated label l_k is defined as

$$AC_{sp}^\lambda(x_i, l_k) = \frac{\sum_{x_t \in sim(x_i)} R_C(x_i, x_t) * SRD(x_i) * R_{l_k}(x_i, x_t)}{|sim(x_i)|} \quad (15)$$

where $R_C(x_i, x_t)$ represents the similarity between sample x_i and x_t in its similar sample set under the feature space, R_{l_k} denotes the similarity between sample x_i and x_t in its similar sample set under the label l_k .

Considering that each label contains different importance levels, we combine the sample label fuzzy association confidence and label weight to obtain the label enhancement value d_i^k of x_i with respect to l_k by noise-tolerant label enhancement.

Definition 14: Given an $MLT = \langle U, C \cup L, V, F \rangle$, $X = \{x_1, x_2, \dots, x_n\}$ is a sample set, $L = \{l_1, l_2, \dots, l_q\}$ is a label set, $LW(l_k)$ represents the importance of label l_k , $AC^\lambda(x_i, l_k)$ denotes the fuzzy association confidence between sample x_i and associated label l_k , the label enhancement value d_i^k of x_i with respect to l_k by noise-tolerant label enhancement is defined as

$$d_i^k = AC^\lambda(x_i, l_k) \cdot LW(l_k) \quad (16)$$

where d_i^k indicates the degree to which label l_k describes the sample x_i by noise-tolerant label enhancement. The noise-tolerant label enhancement value of each instance is normalized by $\tilde{l}_k(x_i) = \frac{d_i^k}{\sum_{l_t \in L} d_i^{l_t}}$ to obtain a noise-tolerant label enhancement information table $NTIT$, which is expressed as $NTIT = \langle U, C \cup \tilde{L} \rangle$, where $\tilde{L} = \{\tilde{l}_1, \tilde{l}_2, \dots, \tilde{l}_q\}$.

The cosine function can be utilized to calculate the similarity between samples with label distribution obtained by noise-tolerant label enhancement under the label space, and the domain of fuzzy similarity relationship is $[0, 1]$. Contrasted with the logical value, the label value obtained by the noise-tolerant label enhancement is more reasonable and reliable. In addition, the fuzzy similarity relation $R_{\tilde{L}}$ satisfies reflexivity and symmetry.

Definition 15: Given a noise-tolerant label enhancement information table $NTIT = \langle U, C \cup \tilde{L} \rangle$, $X = \{x_1, x_2, \dots, x_n\}$ is a sample set, $\tilde{L} = \{\tilde{l}_1, \tilde{l}_2, \dots, \tilde{l}_q\}$ is a noise-tolerant label set, the similarity $R_{\tilde{L}}$ between sample x_i and x_j under the noise-tolerant label enhancement space is defined as

follows:

$$R_{\tilde{L}}(x_i, x_j) = \frac{\sum_{k=1}^q d_i^k d_j^k}{\sqrt{\sum_{k=1}^q (d_i^k)^2} \sqrt{\sum_{k=1}^q (d_j^k)^2}}. \quad (17)$$

C. Noise-Tolerant Fuzzy Confidence Uncertainty Measure

Based on the reliability degree of the sample and the noise-tolerant label space obtained by label enhancement, a new noise-tolerant fuzzy confidence entropy is defined to measure the decision uncertainty of fuzzy granularity in a multilabel information table containing label noise.

Definition 16: Given a noise-tolerant label enhancement information table $NTIT = \langle U, C \cup \tilde{L} \rangle$, $X = \{x_1, x_2, \dots, x_n\}$ is a sample set, $\tilde{L} = \{\tilde{l}_1, \tilde{l}_2, \dots, \tilde{l}_q\}$ is a noise-tolerant label set, $B \subseteq C$, $SRD(x_i)$ indicates the reliability degree of the sample x_i , the detailed definition of noise-tolerant fuzzy confidence entropy under feature subset B for sample x_i is as follows:

$$NTFC_{x_i}(B) = -SRD(x_i) |[x_i]_B \cap [x_i]_{\tilde{L}}| \log \frac{|[x_i]_B|}{n} \quad (18)$$

$$\text{where } [x_i]_{\tilde{L}} = \frac{R_{\tilde{L}}(x_i, x_1)}{x_1} + \frac{R_{\tilde{L}}(x_i, x_2)}{x_2} + \dots + \frac{R_{\tilde{L}}(x_i, x_n)}{x_n}.$$

The noise-tolerant fuzzy confidence entropy of the sample can measure the distinguishing ability of different feature subsets under noise-tolerant label space. Noise-tolerant fuzzy confidence entropy not only takes into account the classification information of the feature space, but also considers a more credible part of the supervisory information obtained by noise-tolerant of the label space. It is easy to see that the distinguishing ability increases with the increase of the cardinality of the feature set.

In order to make the results of MFS more reasonable, we further define the monotonic noise-tolerant fuzzy confidence conditional entropy to measure the uncertainty of labels under known features.

Definition 17: Given a noise-tolerant label enhancement information table $NTIT = \langle U, C \cup \tilde{L} \rangle$, $X = \{x_1, x_2, \dots, x_n\}$ is a sample set, $\tilde{L} = \{\tilde{l}_1, \tilde{l}_2, \dots, \tilde{l}_q\}$ is a noise-tolerant label set, $B \subseteq C$, $SRD(x_i)$ indicates the reliability degree of the sample x_i , the detailed definition of noise-tolerant fuzzy confidence conditional entropy of $\tilde{L}$ in conditional of giving B for sample x_i is as follows:

$$NTFC_{x_i}(\tilde{L}|B) = -SRD(x_i) |[x_i]_B \cap [x_i]_{\tilde{L}}| \log \frac{|[x_i]_B \cap [x_i]_{\tilde{L}}|}{|[x_i]_B|} \quad (19)$$

$$\text{where } [x_i]_{\tilde{L}} = \frac{R_{\tilde{L}}(x_i, x_1)}{x_1} + \frac{R_{\tilde{L}}(x_i, x_2)}{x_2} + \dots + \frac{R_{\tilde{L}}(x_i, x_n)}{x_n}.$$

Proposition 2. (Nonnegative): $NTFC_{x_i}(\tilde{L}|B) \geq 0$.

Proof: By $x_i \in [x_i]_B$ and $x_i \in [x_i]_B \cap [x_i]_{\tilde{L}} (\forall x_i \in U)$, we can easily get $0 < \frac{|[x_i]_B \cap [x_i]_{\tilde{L}}|}{|[x_i]_B|} \leq 1$. Therefore, $NTFC_{x_i}(\tilde{L}|B) \geq 0$ holds. $\square$

Proposition 3. (Monotonicity): Given a noise-tolerant label enhancement information table $NTIT = \langle U, C \cup \tilde{L} \rangle$, $U = \{x_1, x_2, \dots, x_n\}$. For any $x_i \in U$, when $A \subseteq B \subseteq C$, $NTFC_{x_i}(\tilde{L}|B) \leq NTFC_{x_i}(\tilde{L}|A)$ holds.

Proof: For $\forall x_i \in U$, we can get

$$\begin{aligned} |[x_i]_A| &= |[x_i]_A \cap [x_i]_{\tilde{L}}| + |[x_i]_A \cap [x_i]_{\tilde{L}}^c| \\ |[x_i]_B| &= |[x_i]_B \cap [x_i]_{\tilde{L}}| + |[x_i]_B \cap [x_i]_{\tilde{L}}^c| \end{aligned}$$

where $[x_i]_{\tilde{L}}^c = U - [x_i]_{\tilde{L}}$ is the complement of $[x_i]_{\tilde{L}}$. Then if

$$\begin{aligned} |[x_i]_A \cap [x_i]_{\tilde{L}}| &= a_{i1}, |[x_i]_A \cap [x_i]_{\tilde{L}}^c| = a_{i2} \\ |[x_i]_B \cap [x_i]_{\tilde{L}}| &= b_{i1}, |[x_i]_B \cap [x_i]_{\tilde{L}}^c| = b_{i2} \end{aligned}$$

we can obtain the noise-tolerant fuzzy confidence conditional entropy as follows:

$$\begin{aligned} NTFC_{x_i}(\tilde{L}|B) &= -SRD(x_i) b_{i1} \log \frac{b_{i1}}{b_{i1} + b_{i2}} \\ NTFC_{x_i}(\tilde{L}|A) &= -SRD(x_i) a_{i1} \log \frac{a_{i1}}{a_{i1} + a_{i2}}. \end{aligned}$$

Let $F(u, v) = -SRD(x_i) u \log \frac{u}{u+v} (u > 0, v \geq 0)$, and $w = \frac{u}{u+v}$. Then

$$\begin{cases} \frac{\partial F}{\partial u} = -SRD(x_i) \left(\frac{v}{u+v} + \log \frac{u}{u+v} \right) \\ \frac{\partial F}{\partial v} = SRD(x_i) \left(\frac{u}{u+v} \right) > 0. \end{cases}$$

Let $G = \frac{\partial F}{\partial u} = -SRD(x_i) (1 - w + \log w)$, since $SRD(x_i) \geq 0$, then $G' = SRD(x_i) (1 - \frac{1}{w}) \leq 0 (w \in (0, 1])$, we can obtain $\frac{\partial F}{\partial u} \geq 0$. Therefore, the function F increases monotonically with respect to the first and second variables, respectively.

When $A \subseteq B$, $[x_i]_B \subseteq [x_i]_A$ is true, so we can get

$$\begin{aligned} b_{i1} &= |[x_i]_B \cap [x_i]_{\tilde{L}}| \leq |[x_i]_A \cap [x_i]_{\tilde{L}}| = a_{i1} \\ b_{i2} &= |[x_i]_B \cap [x_i]_{\tilde{L}}^c| \leq |[x_i]_A \cap [x_i]_{\tilde{L}}^c| = a_{i2}. \end{aligned}$$

Therefore, $F(b_{i1}, b_{i2}) \leq F(a_{i1}, a_{i2})$, $NTFC_{x_i}(\tilde{L}|B) \leq NTFC_{x_i}(\tilde{L}|A)$ holds. $\square$

D. Proposed SRDAC Algorithm

In summary, for the problem of dimension reduction for noise-tolerant label enhanced multilabel data, a novel robust MFS algorithm based on monotonic noise-tolerant fuzzy confidence conditional entropy is proposed. First, the algorithm converts the logical values of multiple labels in the noise-containing label space into noise-tolerant confidence label enhancement values by label enhancement. Second, for the obtained noise-tolerant confidence multilabel information table, the similarity between instances with noise-tolerant confidence label enhancement values under the label space is computed using cosine similarity to mine the supervised information with real values with higher confidence in the label space. Finally, the importance of candidate features is evaluated based on monotonic noise-tolerant fuzzy confidence conditional entropy.

Definition 18: Given a noise-tolerant label enhancement information table $NTIT = \langle U, C \cup \tilde{L} \rangle$, $S \subseteq C$ represents the selected feature subset, for $\forall f \in C - S$, the importance of feature f relative to S and the noise-tolerant label set $\tilde{L}$ is

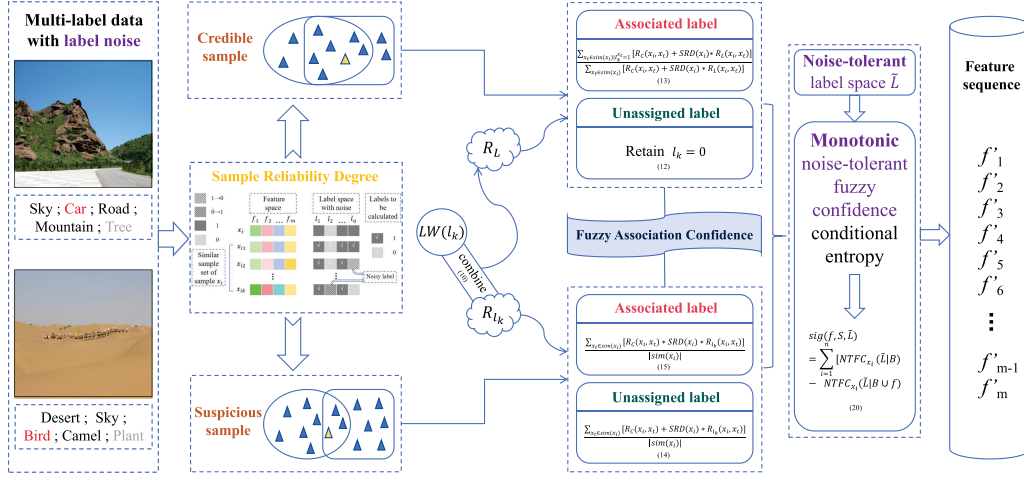


Fig. 3. Framework of our method.

defined as

$$sig(f, S, \tilde{L}) = \sum_{i=1}^n [NTFC_{x_i}(\tilde{L}|B) - NTFC_{x_i}(\tilde{L}|B \cup f)]. \quad (20)$$

The most important candidate feature f_k is added to the subset of selected features at each iteration until a threshold is reached, where f_k can be expressed as

$$f_k = \arg \max_{f \in C-S} \{sig(f, S, \tilde{L})\}. \quad (21)$$

The process of the proposed MFS method (SRDAC) is shown in Algorithm 1. In order to visually explain the process of our method, the framework diagram is illustrated in Fig. 3.

Then, we analyze the time complexity of SRDAC. In multilabel data containing label noise, it is assumed that m is the number of features, n is the number of samples, and q is the number of labels, g represents the number of the selected features. SRDAC is mainly divided into two parts: noise-tolerant label enhancement and MFS. Among them, the time complexity of noise-tolerant label enhancement can be considered as $O(nq + q^2)$. In addition, the time complexity of feature selection process is $O(nmg)$. To summarize, the total time complexity of SRDAC is about $O(nq + q^2 + nmg)$.

IV. EXPERIMENTAL ANALYSIS

In this section, a series of experiments between SRDAC and other representative methods are conducted to verify the effectiveness and robustness of SRDAC, and the obtained experimental results are presented and analyzed.

A. Description of Experimental Datasets

To verify the effectiveness of SRDAC, 17 multilabel datasets downloaded from Mulan Library [31] are selected for a series of experiments in this article. Table I lists some information about these 17 datasets, including sample size (Ins), number of

TABLE I
INFORMATION ABOUT EXPERIMENTAL DATASETS

No.	Datasets	Ins	Training	Test	Features	Labels	Card	Dens	Dom
1	Arts	5000	2000	3000	462	26	1.636	0.063	Text
2	Birds	645	322	323	260	19	1.014	0.053	Audio
3	Business	5000	2000	3000	438	30	1.588	0.053	Text
4	Computers	5000	2000	3000	681	33	1.508	0.046	Yahoo
5	Emotions	593	391	202	72	6	1.869	0.311	Music
6	Scene	2407	1211	1196	294	6	1.074	0.179	Image
7	Medical	978	333	645	1449	45	1.245	0.028	Text
8	Yeast	2417	1500	917	103	14	4.237	0.303	Biology
9	Flags	194	129	65	19	7	3.392	0.485	Image
10	Entertainment	5000	2000	3000	640	21	1.420	0.068	Text
11	Education	5000	2000	3000	550	33	1.461	0.044	Text
12	Health	5000	2000	3000	612	32	1.662	0.052	Text
13	Recreation	5000	2000	3000	606	22	1.423	0.065	Yahoo
14	Reference	5000	2000	3000	793	33	1.169	0.035	Yahoo
15	Society	5000	2000	3000	636	27	1.692	0.063	Text
16	EukaryotePseAAC	7766	3107	4659	440	22	1.146	0.052	Biology
17	Yelp	10806	3242	7564	671	5	1.638	0.328	Text

training and test sets, number of features, number of labels, label cardinality (Card), label density (Dens), and domains (Dom).

B. Experiment Settings

In order to evaluate the performance of SRDAC, we compare SRDAC with seven other representative MFS algorithms. Brief descriptions of these comparison algorithms are given below.

SCLS [32]: An MFS method based on scalable relevance evaluation. MDFS [4]: An embedded MFS method based on manifold regularization. MCDM [33]: An MFS method using multicriteria decision making. VMFS [34]: A multitarget feature selection method based on VIKOR. GLFS [35]: An MFS that integrates global and local features. NRFSFN [23] and GRMFS [24] have been introduced in introduction.

The parameters of each selected comparison algorithms are set according to recommendations of the corresponding literature. Specifically, for MCDM, parameter λ is set to 10 according to the recommendation in [33]. SCLS and VMFS have no parameters. For GLFS, β is searched in $\{10^{-3}, 10^{-2}, \dots, 10^3\}$. For MDFS, γ is searched in $\{10^{-3}, 10^{-2}, \dots, 10^3\}$. For NRFSFN, α is searched in $[0.5, 1]$, with a step size of 0.1. For GRMFS, k is searched in $[1, 5]$, with a step size of 1. For SRDAC, when

Algorithm 1: Noise-Tolerant Label Feature Selection Based on Sample Reliability Degree and Fuzzy Association Confidence(SRDAC).

Input: Multilabel information table

$MLT = \langle U, C \cup L, V, F \rangle$, the confidence threshold λ , and the proportion of features selected $numK$.

Output: A reduced feature subset S .

- 1: Calculate the label weight $LW(l_k)$ according to (4);
- 2: **for** each $f \in C$ **do**
- 3: Compute the fuzzy similarity relationship between samples in the feature space;
- 4: **end for**
- 5: Calculate the reliability degree of each sample $SRD(x_i)$ according to (6)–(7);
- 6: Divide the samples into credible samples and suspicious samples according to (8)–(9);
- 7: Calculate the similarity under the entire original label set L according to (10);
- 8: Calculate the fuzzy association confidence between different types of samples and corresponding labels according to (12)–(15);
- 9: Combined with the label weight $LW(l_k)$, calculate the noise-tolerant label enhancement value according to (16);
- 10: $S \leftarrow \emptyset$;
- 11: Calculate the similarity between samples under the noise-tolerant label enhancement space according to (17);
- 12: $S \leftarrow$ find the best candidate feature a_k based on (19);
- 13: **while** $|S| \leq numK$ **do**
- 14: **for** each $f \in C - S$ **do**
- 15: Calculate the importance of feature f according to (20);
- 16: $f_s \leftarrow$ find the best candidate feature f_k ;
- 17: $S \leftarrow S \cup f_s$;
- 18: **end for**
- 19: **end while**
- 20: **return** S .

the number of training samples is less than 1000, the value of k in (6) is taken as 10. When the number of training samples is greater than or equal to 1000, the value of k in (6) is taken as 50. λ is searched in $[0.1, 0.7]$. In this article, we use MLKNN as a multilabel classifier to evaluate the performance of algorithms. The smoothing parameter is set to 1 and the nearest neighbor is set to 10 [36]. We select five commonly used multilabel learning evaluation metrics, including average precision (AP), one error (OE), ranking loss (RL), coverage (CV), and Macro-F1 (Mac-F1) [37], to evaluate classification performance of all algorithms. “ $\uparrow$ ” indicates that a larger value is better, and “ $\downarrow$ ” indicates that a smaller value is better. We simulate label noise by replacing positive label (+1) with negative label (−1) at a randomly chosen ratio from 0% to 40% in the label space, while replacing negative label (−1) with positive label (+1) in the same ratio. Since the above algorithms can all obtain an ordered

TABLE II
PERFORMANCE COMPARISON OF ALL ALGORITHMS ON AP METRIC ($\uparrow$)

Datasets	SCLS	MDFS	MCDM	VMFS	GLFS	NRF5FN	GRMFS	SRDAC
Arts	0.4742	0.4729	0.4851	0.4680	0.4837	0.4830	0.4869	0.4931
Birds	0.6338	0.6434	0.6427	0.6783	0.6784	0.6679	0.6930	0.6986
Business	0.8609	0.8656	0.8633	0.8651	0.8673	0.8672	0.8678	0.8717
Computers	0.6192	0.6194	0.6134	0.6234	0.6175	0.6184	0.6250	0.6295
Emotions	0.6258	0.7282	0.7324	0.6219	0.7141	0.7164	0.7157	0.6985
Scene	0.7572	0.7950	0.7765	0.6960	0.7766	0.6942	0.7450	0.8049
Medical	0.7336	0.4803	0.5921	0.4769	0.7484	0.7397	0.7368	0.7747
Yeast	0.7321	0.7316	0.7302	0.7356	0.7388	0.7276	0.7301	0.7433
Flags	0.7957	0.7949	0.7946	0.7981	0.7834	0.7940	0.7916	0.8039
Entertainment	0.5529	0.5456	0.5104	0.5476	0.5565	0.5501	0.5502	0.5601
Education	0.4948	0.5186	0.5159	0.5142	0.5294	0.5307	0.5367	0.5462
Health	0.6400	0.6489	0.6355	0.6567	0.6700	0.6647	0.6584	0.6739
Recreation	0.4814	0.4377	0.4200	0.4241	0.4365	0.4418	0.4556	0.4718
Reference	0.6020	0.6074	0.5759	0.6031	0.5855	0.5918	0.6093	0.6189
Society	0.5585	0.5669	0.5520	0.5662	0.5598	0.5572	0.5743	0.5854
EukaryotePseAAC	0.4976	0.5520	0.5107	0.5428	0.5173	0.5247	0.5285	0.5346
Yelp	0.7534	0.7481	0.7534	0.7488	0.7483	0.7510	0.7493	0.7554
Average	0.6361	0.6327	0.6297	0.6216	0.6477	0.6424	0.6502	0.6626
Rank.Avg	5.3824	4.7059	6.0294	5.2941	4.4706	5.0000	3.6471	1.4706

TABLE III
PERFORMANCE COMPARISON OF ALL ALGORITHMS ON OE METRIC ($\downarrow$)

Datasets	SCLS	MDFS	MCDM	VMFS	GLFS	NRF5FN	GRMFS	SRDAC
Arts	0.6823	0.6845	0.6541	0.6943	0.6689	0.6716	0.6653	0.6521
Birds	0.4848	0.4570	0.4566	0.3981	0.3894	0.4149	0.3680	0.3634
Business	0.1368	0.1338	0.1336	0.1359	0.1336	0.1335	0.1326	0.1288
Computers	0.4557	0.4568	0.4707	0.4549	0.4617	0.4614	0.4536	0.4448
Emotions	0.4988	0.3824	0.3822	0.5045	0.4057	0.4072	0.4079	0.3839
Scene	0.3894	0.3346	0.3584	0.4700	0.3646	0.4758	0.4112	0.3118
Medical	0.3329	0.6385	0.4752	0.6490	0.3229	0.3367	0.3369	0.2794
Yeast	0.2506	0.2502	0.2555	0.2513	0.2457	0.2531	0.2467	0.2408
Flags	0.2338	0.2308	0.2346	0.2269	0.2581	0.2358	0.2277	0.2027
Entertainment	0.6114	0.6140	0.6459	0.6105	0.5899	0.6076	0.6049	0.5912
Education	0.6666	0.6322	0.6320	0.6383	0.6211	0.6155	0.6079	0.5948
Health	0.4617	0.4476	0.4646	0.4455	0.4188	0.4378	0.4431	0.4200
Recreation	0.6601	0.7243	0.7435	0.7437	0.7293	0.7202	0.7028	0.6766
Reference	0.4866	0.4834	0.5246	0.5013	0.5160	0.5101	0.4931	0.4793
Society	0.4863	0.4828	0.4999	0.4858	0.4911	0.4950	0.4736	0.4585
EukaryotePseAAC	0.6969	0.6258	0.6794	0.6405	0.6724	0.6603	0.6564	0.6482
Yelp	0.3631	0.3653	0.3648	0.3650	0.3638	0.3646	0.3649	0.3610
Average	0.4646	0.4673	0.4692	0.4833	0.4502	0.4589	0.4469	0.4257
Rank.Avg	5.3529	4.8235	5.9118	5.6471	4.2647	5.0588	3.5294	1.4118

TABLE IV
PERFORMANCE COMPARISON OF ALL ALGORITHMS ON RL METRIC ($\downarrow$)

Datasets	SCLS	MDFS	MCDM	VMFS	GLFS	NRF5FN	GRMFS	SRDAC
Arts	0.1618	0.1628	0.1663	0.1609	0.1602	0.1570	0.1554	0.1553
Birds	0.1571	0.1565	0.1528	0.1367	0.1406	0.1371	0.1330	0.1307
Business	0.0483	0.0448	0.0474	0.0453	0.0447	0.0444	0.0448	0.0432
Computers	0.0994	0.0968	0.0973	0.0960	0.0971	0.0957	0.0957	0.0942
Emotions	0.3774	0.2331	0.2333	0.3840	0.2586	0.2459	0.2467	0.2933
Scene	0.1565	0.1264	0.1447	0.2137	0.1387	0.2115	0.1639	0.1265
Medical	0.0672	0.1179	0.1016	0.1165	0.0597	0.0603	0.0607	0.0580
Yeast	0.1909	0.1916	0.1932	0.1890	0.1872	0.1948	0.1934	0.1849
Flags	0.2411	0.2377	0.2445	0.2412	0.2408	0.2418	0.2452	0.2296
Entertainment	0.1282	0.1292	0.1443	0.1286	0.1259	0.1284	0.1278	0.1258
Education	0.1034	0.0991	0.1011	0.0995	0.0968	0.0967	0.0940	0.0923
Health	0.0745	0.0725	0.0757	0.0676	0.0675	0.0648	0.0669	0.0641
Recreation	0.1899	0.2005	0.2092	0.2017	0.1999	0.1965	0.1906	0.1882
Reference	0.0959	0.0944	0.0997	0.0897	0.0980	0.0950	0.0884	0.0874
Society	0.1575	0.1545	0.1590	0.1511	0.1550	0.1555	0.1500	0.1468
EukaryotePseAAC	0.1430	0.1295	0.1402	0.1312	0.1383	0.1371	0.1352	0.1348
Yelp	0.2343	0.2404	0.2326	0.2399	0.2406	0.2365	0.2392	0.2321
Average	0.1545	0.1463	0.1496	0.1584	0.1441	0.1470	0.1430	0.1404
Rank.Avg	5.5714	4.7353	6.4118	5.1176	4.3529	4.2647	3.7647	1.4706

feature set, the selection proportion for features is uniformly fixed at 40% of the total features.

C. Comparison on Classification Performance

The classification results of SRDAC and other algorithms across five evaluation metrics are presented in Tables II–VI, where the best results are in bold. Among them, the values reflect the average performance for feature selection proportions from 1% to 40% in each dataset. Average is utilized to indicate the average performance of a single algorithm across all datasets. Rank.Avg (Rank Average) indicates the average ranking of a single algorithm on all datasets, and a smaller value indicates

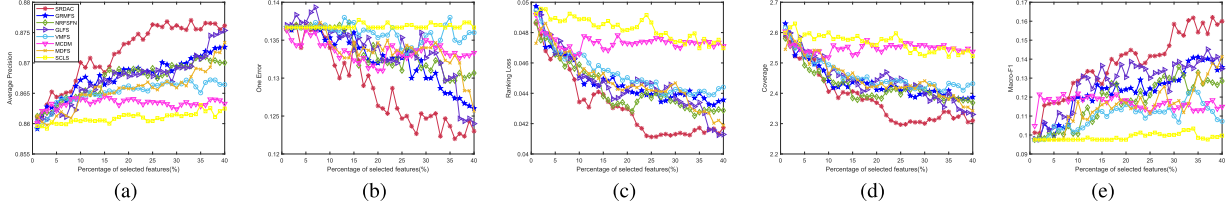


Fig. 4. Performance comparison experiments with different feature ratios on the business dataset. (a) AP. (b) OE. (c) RL. (d) CV. (e) Mac-F1.

TABLE V
PERFORMANCE COMPARISON OF ALL ALGORITHMS ON CV METRIC (↓)

Datasets	SCLS	MDFS	MCDM	VMFS	GLFS	NRFSGN	GRMFS	SRDAC
Arts	5.7262	5.7399	5.8509	5.6689	5.6908	5.5922	5.5399	5.5373
Birds	4.0739	4.0803	4.0258	3.6162	3.7824	3.6294	3.6128	3.5423
Business	2.5668	2.4484	2.5493	2.4602	2.4456	2.4259	2.4384	2.3826
Computers	4.7117	4.6168	4.6173	4.5836	4.6306	4.5581	4.5697	4.5109
Emotions	2.9890	2.2434	2.2454	3.0225	2.3625	2.3067	2.3031	2.5536
Scene	0.8869	0.7371	0.8241	1.1744	0.7977	1.1622	0.9245	0.7335
Medical	3.8348	6.1736	5.4733	6.1025	3.4912	3.5693	3.5740	3.4400
Yeast	6.6754	6.6624	6.7119	6.6284	6.6045	6.7378	6.7019	6.6232
Flags	3.9462	3.8796	3.8454	3.8988	3.8977	3.9319	3.9350	3.8923
Entertainment	3.4071	3.4361	3.7740	3.4208	3.3753	3.4207	3.4124	3.3643
Education	4.3168	4.1606	4.2175	4.1817	4.0997	4.0827	3.9925	3.9373
Health	3.8124	3.7423	3.8518	3.5657	3.5808	3.4481	3.5341	3.4334
Recreation	5.0592	5.2577	5.4356	5.2691	5.2617	5.1644	5.0527	5.0209
Reference	3.6756	3.6296	3.8017	3.4596	3.7439	3.6332	3.4156	3.3878
Society	6.1002	5.9953	6.1188	5.8965	6.0173	6.0253	5.8556	5.7755
EukaryotePseAAC	3.2743	2.9936	3.2161	3.0296	3.1719	3.1522	3.1165	3.1026
Yelp	1.5109	1.5433	1.5090	1.5420	1.5451	1.5259	1.5375	1.5081
Average	3.9157	3.9612	4.0040	3.9718	3.7940	3.7862	3.7362	3.6909
Rank.Avg	6.1176	4.7647	6.0588	5.0000	4.6471	4.2941	3.5294	1.5882

TABLE VI
PERFORMANCE COMPARISON OF ALL ALGORITHMS ON MACRO-F1 METRIC (↑)

Datasets	SCLS	MDFS	MCDM	VMFS	GLFS	NRFSGN	GRMFS	SRDAC
Arts	0.0332	0.0347	0.0489	0.0275	0.0393	0.0306	0.0475	0.0518
Birds	0.0571	0.0644	0.0706	0.1133	0.0913	0.0933	0.1225	0.1382
Business	0.0989	0.1152	0.1169	0.1099	0.1277	0.1177	0.1228	0.1381
Computers	0.0464	0.0451	0.0400	0.0510	0.0557	0.0515	0.0524	0.0706
Emotions	0.2073	0.4453	0.4395	0.1693	0.4397	0.4086	0.4400	0.3197
Scene	0.5110	0.5737	0.5473	0.4071	0.5426	0.3586	0.4655	0.6241
Medical	0.2566	0.1596	0.2036	0.1532	0.2747	0.2639	0.2588	0.2761
Yeast	0.2795	0.2811	0.2636	0.2986	0.2988	0.2639	0.2605	0.3092
Flags	0.4418	0.4165	0.5279	0.4753	0.4222	0.4582	0.4452	0.4671
Entertainment	0.0810	0.0792	0.0596	0.0806	0.1104	0.0736	0.0859	0.1009
Education	0.0793	0.0953	0.0982	0.0887	0.0945	0.1055	0.1113	0.1209
Health	0.1122	0.1383	0.1227	0.1704	0.1733	0.1911	0.1719	0.1949
Recreation	0.0893	0.0402	0.0344	0.0323	0.0523	0.0441	0.0646	0.0807
Reference	0.1130	0.1100	0.0852	0.0992	0.0869	0.0926	0.1068	0.1168
Society	0.0292	0.0376	0.0195	0.0331	0.0291	0.0280	0.0464	0.0579
EukaryotePseAAC	0.0045	0.0212	0.0044	0.0169	0.0106	0.0127	0.0124	0.0176
Yelp	0.2362	0.1779	0.2290	0.1698	0.1782	0.1910	0.1657	0.2179
Average	0.1574	0.1668	0.1713	0.1468	0.1781	0.1638	0.1753	0.1943
Rank.Avg	5.4118	4.8824	5.5294	5.5882	4.0588	4.8824	3.9412	1.7059

better performance. According to Tables II–VI, we can draw the following conclusions:

On AP metric, SRDAC leads on 14 datasets. In addition, SRDAC ranks second on Recreation, just behind SCLS. For OE metric, SRDAC performs superior on 12 datasets and obtains suboptimal results for entertainment, health, and recreation. Regarding the RL metric, SRDAC outperforms other comparison algorithms on 14 datasets and wins the second place on Scene. As can be seen from the CV metric, SRDAC performs well on 13 datasets and achieves suboptimal performance on Yeast. As shown by the Macro-F1 metric in Table VI, SRDAC achieves competitive performance on 11 datasets. From the average of the classification results of five evaluation metrics, SRDAC has obvious advantages. Therefore, SRDAC achieves superior performance over other algorithms.

To further verify the accuracy and stability of SRDAC, Figs. 4 and 5 show the performance variation of all algorithms under

different evaluation metrics in the range of the proportion of selected features on Business and Society from 1% to 40%, with a step size of 1%. The horizontal axis represents the proportion of selected features, and the vertical axis represents the performance on different metrics. As can be seen from Figs. 4 and 5, as the percentage of selected features increases, SRDAC has obvious advantages over other algorithms, with both optimal and average values exceeding those of other algorithms, and with higher stability. Therefore, SRDAC is superior to other algorithms.

D. Verification of Robustness

In order to prove the robustness of SRDAC, as shown in Figs. 6–9, the classification performance of five evaluation metrics of SRDAC and other algorithms under different label noise ratios is demonstrated on Education, Health, Medical, and Scene. As can be seen from Figs. 6–9, as label noise ratio changes, SRDAC fluctuates less compared to other algorithms and has outstanding classification performance, obtaining optimal performance at almost all noise ratios. Specifically, we draw the following conclusions.

For the education dataset, SRDAC has obvious advantages in performance of five metrics, while the performance of SCLS fluctuates less. From five metrics of the health dataset, SRDAC is in a leading position and has higher stability under different noise ratios. Regarding the medical dataset, the performance of SRDAC is superior to other algorithms. VMFS has less performance fluctuation, but its overall performance is not outstanding. In the scene dataset, SRDAC achieves superior classification performance relative to other algorithms.

In general, compared with other representative algorithms, SRDAC can achieve superior classification performance while maintaining competitive robustness.

E. Statistical Analysis

In order to systematically analyze the performance difference between SRDAC and the other algorithms, we adopt Friedman test [38] and Bonferroni–Dunn test [39] to verify the validity of SRDAC from the angle of statistics. Assuming that there are N datasets and k methods, r_i^j denotes the ranking of the i th method on the j th dataset. $R_i = \frac{1}{N} \sum_{j=1}^N r_i^j$ denotes the average ranking of the i th method on all datasets. Therefore, the chi-square is defined as

$$\chi_F^2 = \frac{12 * N}{k * (k + 1)} * \left(\sum_{i=1}^k R_i^2 - \frac{k * (k + 1)^2}{4} \right) \quad (22)$$

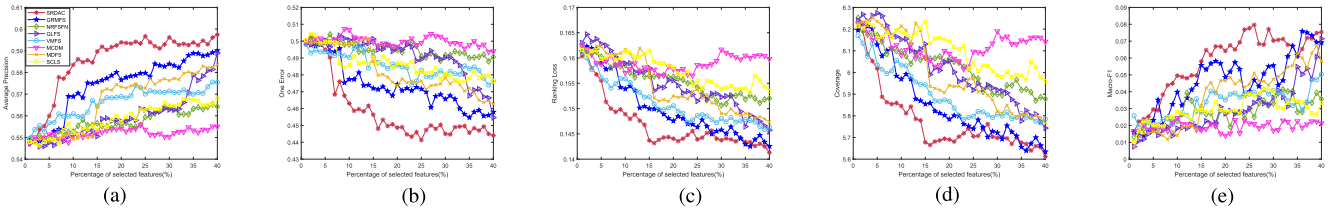


Fig. 5. Performance comparison experiments with different feature ratios on the society dataset. (a) AP. (b) OE. (c) RL. (d) CV. (e) Mac-F1.

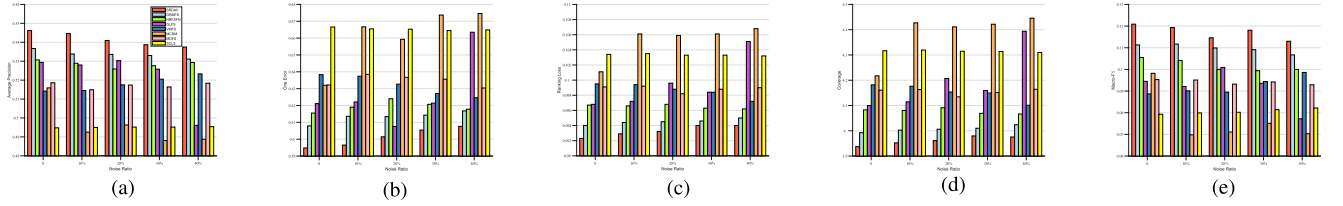


Fig. 6. Robustness experiments on the education dataset under different ratios of label noise. (a) AP. (b) OE. (c) RL. (d) CV. (e) Mac-F1.

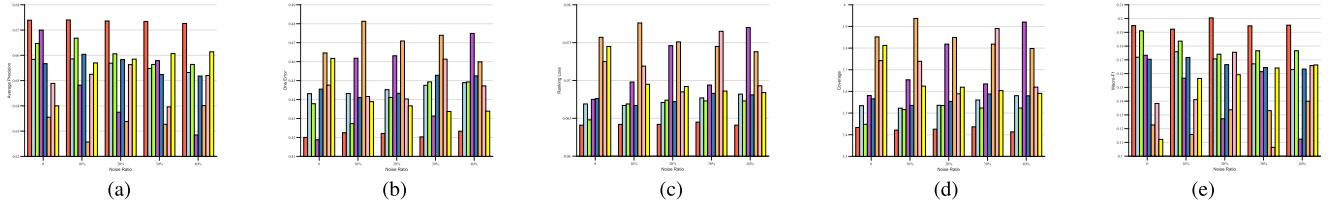


Fig. 7. Robustness experiments on the health dataset under different ratios of label noise. (a) AP. (b) OE. (c) RL. (d) CV. (e) Mac-F1.

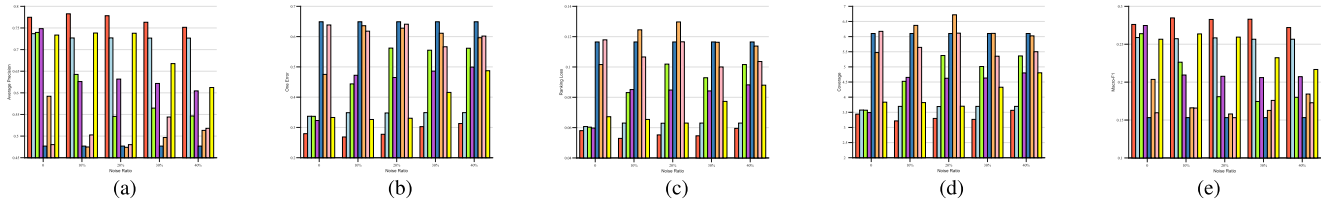


Fig. 8. Robustness experiments on the medical dataset under different ratios of label noise. (a) AP. (b) OE. (c) RL. (d) CV. (e) Mac-F1.

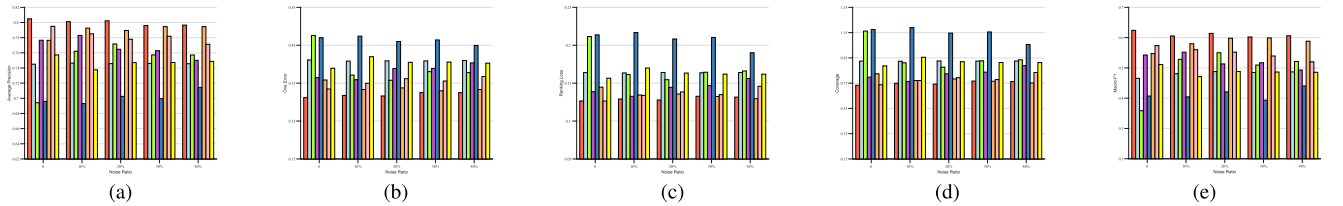


Fig. 9. Robustness experiments on the scene dataset under different ratios of label noise. (a) AP. (b) OE. (c) RL. (d) CV. (e) Mac-F1.

where the degree of freedom of χ_F^2 is $(k - 1)$. The null hypothesis is that all methods have the same classification performance. The Friedman statistical formula for chi-square χ_F^2 is expressed as

$$F_F = \frac{(N - 1) * \chi_F^2}{N(k - 1) - \chi_F^2} \quad (23)$$

where F_F is the F distribution based on degrees of freedom $(k - 1)(N - 1)$ and $(k - 1)$.

By setting the importance level $\alpha = 0.05$, we can obtain a critical value of 2.0924. Table VII shows the Friedman test results of five evaluation metrics. Since Friedman statistics of five evaluation metrics are much larger than the critical value, the

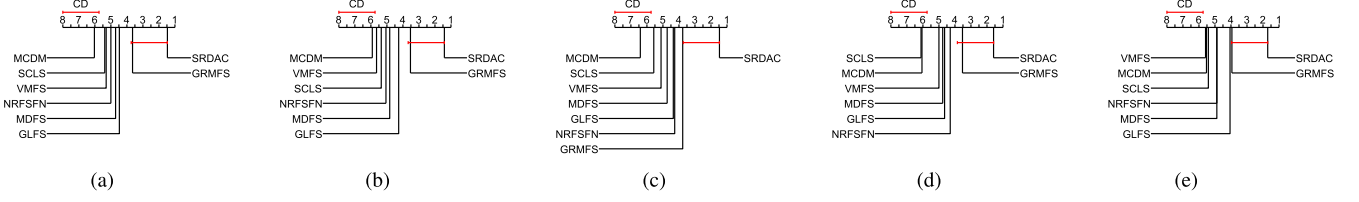
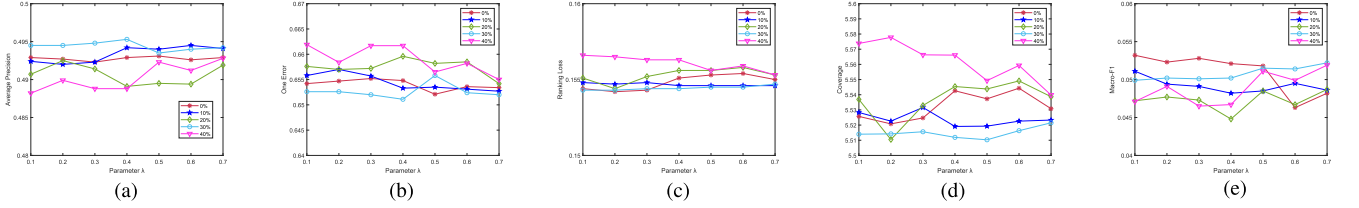


Fig. 10. CD plots. (a) AP. (b) OE. (c) RL. (d) CV. (e) Macro-F1.

Fig. 11. Classification performance of SRDAC on the arts dataset under different label noise ratios varies with the parameter λ . (a) AP. (b) OE. (c) RL. (d) CV. (e) Macro-F1.TABLE VII
FRIEDMAN TEST UNDER ALL METRICS

Evaluation metrics	F_F	Critical value
Average precision	7.9548	2.0924
One error	8.8776	
Ranking loss	6.5773	
Coverage	8.7517	
Macro-F1	6.1654	

null hypothesis is rejected, i.e., there is a significant difference in the performance of all algorithms.

Next, we chose the Bonferroni–Dunn test as a follow-up test to verify significant differences between SRDAC and other methods. The critical difference (CD) can be utilized as a benchmark for the Bonferroni–Dunn test. The formula for CD is as follows:

$$CD_{\alpha} = \rho_{\alpha} \sqrt{\frac{k * (k + 1)}{6 * N}} \quad (24)$$

where α is set to 0.05, and $CD = 2.2601$ can be obtained at $k = 8, N = 17$.

If the difference between the average ranks of the comparison method and SRDAC falls within a CD, they are considered to have statistically similar classification performance. On the contrary, it is considered that the performance of this method is significantly different from that of SRDAC. Fig. 10 shows the CD plots of five evaluation metrics, with a 1-D axis illustrating the average ranking of each method. The red line denotes the CD interval between SRDAC and other methods. The following conclusions can be drawn from Fig. 10: SRDAC ranks the best under every evaluation metric. In AP, OE, RL, and CV, SRDAC is significantly superior to GLFS, VMFS, MCDM, SCLS, MDIFS, and NRFSFN methods. For the Macro-F1 metric, SRDAC is significantly superior to VMFS, MCDM, SCLS, MDIFS, and NRFSFN methods. Although GRMFS is all in the CD range of SRDAC, we can see that there is still a certain performance gap with SRDAC. Overall, SRDAC is significantly better than other comparison methods in all evaluation metrics.

F. Parameter Sensitivity Analysis

SRDAC has one parameter λ . $\lambda \in [0.1, 0.7]$ determines the classification result of credible samples and suspicious samples, and the step size of λ is 0.1. We investigate the variation of classification performance with λ at different noise ratios on Arts. Fig. 11 shows the classification results of Arts on five metrics: AP, OE, RL, CV, and Macro-F1. The horizontal axis represents the different values of λ , and the vertical axis represents the classification performance. The different colored polylines represent the five different proportions of label noise.

The following conclusions can be drawn from Fig. 11. Different λ will have different impacts on classification results of SRDAC. When label space contains 40% noise, the performance of Arts on CV metric fluctuates to some extent. In addition, SRDAC shows relatively stable performance on five metrics.

Therefore, we can obtain the optimal classification performance of SRDAC on each dataset by traversing different values of λ . For example, in a label space with 30% noise, the optimal performance of SRDAC on the OE metric is obtained at $\lambda = 0.4$.

V. CONCLUSION

MFS is an effective data preprocessing technique to improve classification performance by removing redundant and irrelevant features. In practical applications, due to the inevitable error of data labeling and the ambiguity of multilabel space, the obtained data usually contained label noise. Considering that the existing MFS methods for label noise ignored the fact that the label sets of different samples were affected by noise to different degrees and the ambiguity between samples and labels, this article proposed a novel noise-tolerant MFS method based on SRD and fuzzy association confidence. First, the SRD was proposed to divide samples into credible samples and suspicious samples. Second, the fuzzy association confidence measure between samples and labels was proposed from the perspective of global similarity to carry out different noise-tolerant label enhancement for different

types of samples, which reduced the interference caused by label noise. Third, unlike fuzzy conditional entropy that does not have monotonicity, the noise-tolerant fuzzy confidence conditional entropy was proposed. Based on the monotonic noise-tolerant fuzzy confidence conditional entropy, a novel robust MFS algorithm was proposed to improve the classification performance of multilabel data whose label space is affected by noise. Finally, a large number of experimental results showed that SRDAC could obtain superior classification performance while maintaining competitive robustness compared with other representative methods. Since multilabel data acquired in the real world may not be able to obtain all features or labels at once, but gradually over time, we will further study the MFS methods for streaming features and streaming labels in the future.

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